Rishit Yogesh Bafna

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Education

Arizona State University — Tempe, AZ
Master of Science in Computer Science — *Jan 2026 – May 2027*

Arizona State University — Tempe, AZ
Bachelor of Science in Computer Science (Software Engineering) – GPA: 3.42 — *Dec 2025*

- New American University Scholar (4-year merit scholarship), Dean's List (multiple semesters)
- Relevant Coursework: Operating Systems, Distributed Systems, Machine Learning, Data Structures and Algorithms, Software Design

Projects

RiskStream | *Python, XGBoost, FastAPI, PostgreSQL, Redis, MLflow, Docker* — *Mar 2026 – May 2026*

- Engineered end-to-end fraud detection pipeline over 500K transactions — point-in-time correct features eliminate label leakage with training/serving parity verified by automated tests.
- Achieved PR-AUC of 0.96 on 0.17% class imbalance using XGBoost + Isolation Forest with Platt calibration; inference serves at p99 <50ms with zero database reads on the hot path.
- Detects distribution shift and self-retrains automatically — PSI-based drift monitoring triggers model retraining and hot-swaps the new version without service restart.

CyberSentient RAG Pipeline | *Python, Qwen3-Embedding-8B, Qdrant, BM25, FastAPI* — *Jan 2026 – May 2026*

- Designed hybrid retrieval pipeline over 100K+ CVE/CWE/CAPEC records combining dense vector search, BM25 sparse retrieval, and Reciprocal Rank Fusion.
- Applied cross-encoder reranking (Qwen3-Reranker-8B) to disambiguate semantically similar threat queries, reducing false retrievals on ambiguous CVE lookups.
- Achieved 400ms end-to-end query latency for real-time cybersecurity threat intelligence across a 6-feed ingestion pipeline.

Distributed Key-Value Store | *Java, Raft Consensus, gRPC, Protocol Buffers* — *Oct 2025 – Jan 2026*

- Implemented fault-tolerant distributed key-value store using Raft consensus with leader election and log replication across a 5-node cluster.
- Designed gRPC API for SET/GET/DELETE operations achieving sub-5ms latency with correct leader redirection.
- Validated fault tolerance via chaos testing — simulated leader failures and verified automatic recovery with zero data loss.

Experience

Software Engineering Intern — Aug 2025 – Dec 2025
J. Miller Custom Cues — *Tempe, AZ*

- Shipped production 3D product configurator using Three.js and React, reducing customer design revision cycles by 35%.
- Built PostgreSQL REST API with Stripe integration processing 100+ orders with transactional cart management and sub-200ms response times.

Software Engineering Intern — May 2025 – Jul 2025
Winssoft Technologies India Pvt. Ltd. — *Mumbai, India*

- Optimized SQL queries over 200K+ records using composite indexes and materialized views, reducing dashboard load time from 8s to 2s.
- Refactored payment microservice handling 5K+ daily transactions with idempotency guarantees, preventing duplicate charge errors in concurrent request scenarios.

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs, gRPC, WebSockets
Databases & Tools: PostgreSQL, Redis, MongoDB, MLflow, Docker, Git, Linux, AWS (S3, EC2)
Machine Learning: XGBoost, scikit-learn, RAG Pipelines, Vector Databases (Qdrant)
Frontend: React, Three.js, Next.js, Tailwind CSS